CRC Cards
System Analysis and Design

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What are CRC cards?

- CRC = Class-Responsibility-Collaboration
- Represented in the form of cards
- The cards are divided into 3 portions - Class, Responsibility and Collaboration.

Class

- Just like the object oriented programming concept, a class is a collection of similar items.
- A template for object creation.
- An object is a place, thing, event or concept relevant to the system.
- E.g. in a University system, classes can be students, teachers or seminars etc.
- They often resemble (but are not limited to) the actors in a system.
- Usually presented in singular form.

Responsibility

- A responsibility is something that a class either knows or does/performs.
- For example, customers have names, addresses etc. These are what a customer knows or has.
- Besides, customers do the following: they place orders, give feedback etc. These are what a customer does.
- So all these are responsibilities of the class Customer: what they know and also what they do.
- Note: A class can change the value of what it knows but cannot change the values of what other classes know.

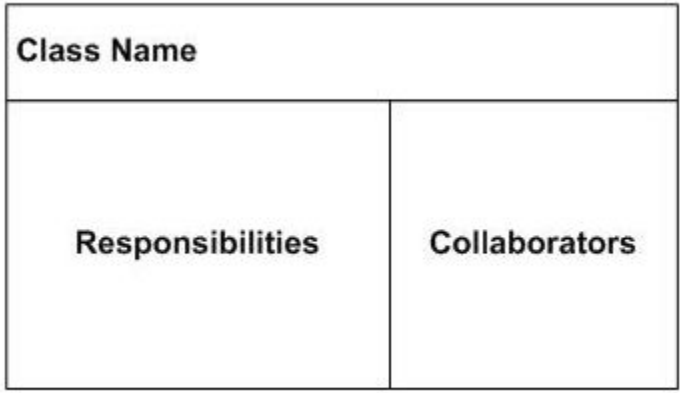
Collaboration

- Sometimes a class needs to depend on other classes.
- It may have a responsibility to fulfill but not enough information to do it.
- For example, a student may want to enroll in a seminar. But for that, the student needs to know if a spot is available in the seminar. But they might not have information about them.
- So the Student class needs to collaborate with the Seminar class.
- Collaboration or interaction occurs between multiple classes.
- It can be a request for information or a request to do something.

Format

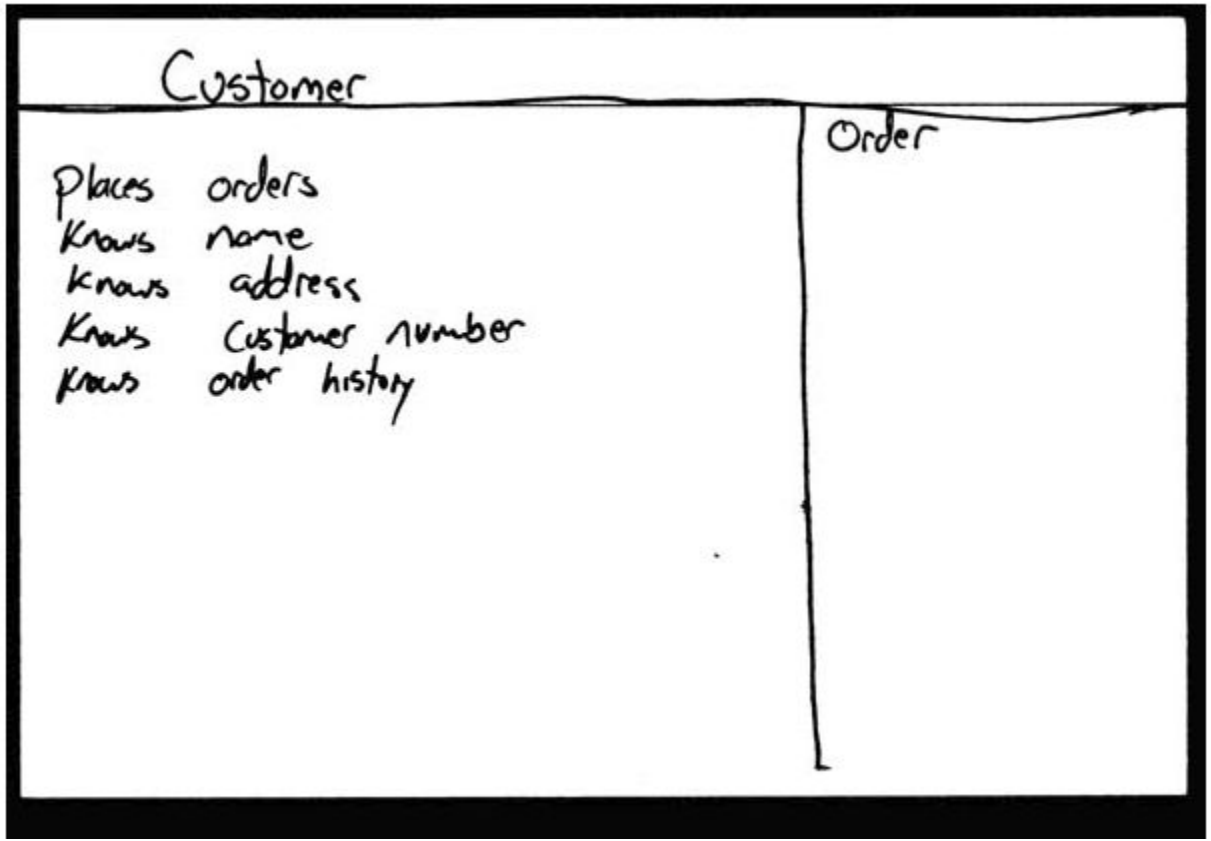
- On top of the card, we have to write the class name.
- On the left, we will have to write the responsibilities.
- On the right, we will have to write the collaborators of the class (with which the class interacts to fulfill its responsibilities).

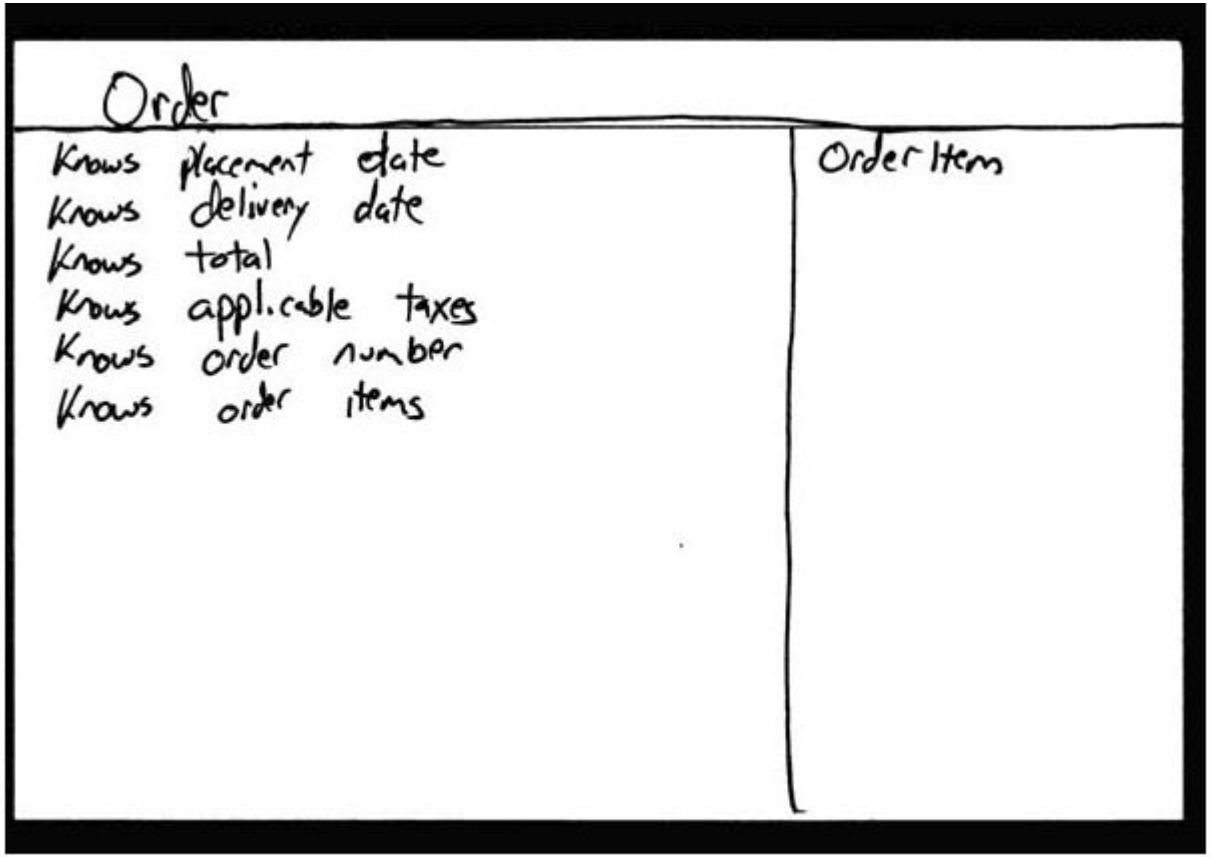
Format



How to brainstorm CRC cards?

- Brainstorm to find the actors in a system and what they do.
- Nouns can be translated to Classes.
- Verbs can be translated to responsibilities.
- Other classes that the class interacts with can be collaborators.
- The class attributes can also be responsibilities.


Customer
Order
Places orders
Knows name
Knows address
Knows Customer number
Knows order history

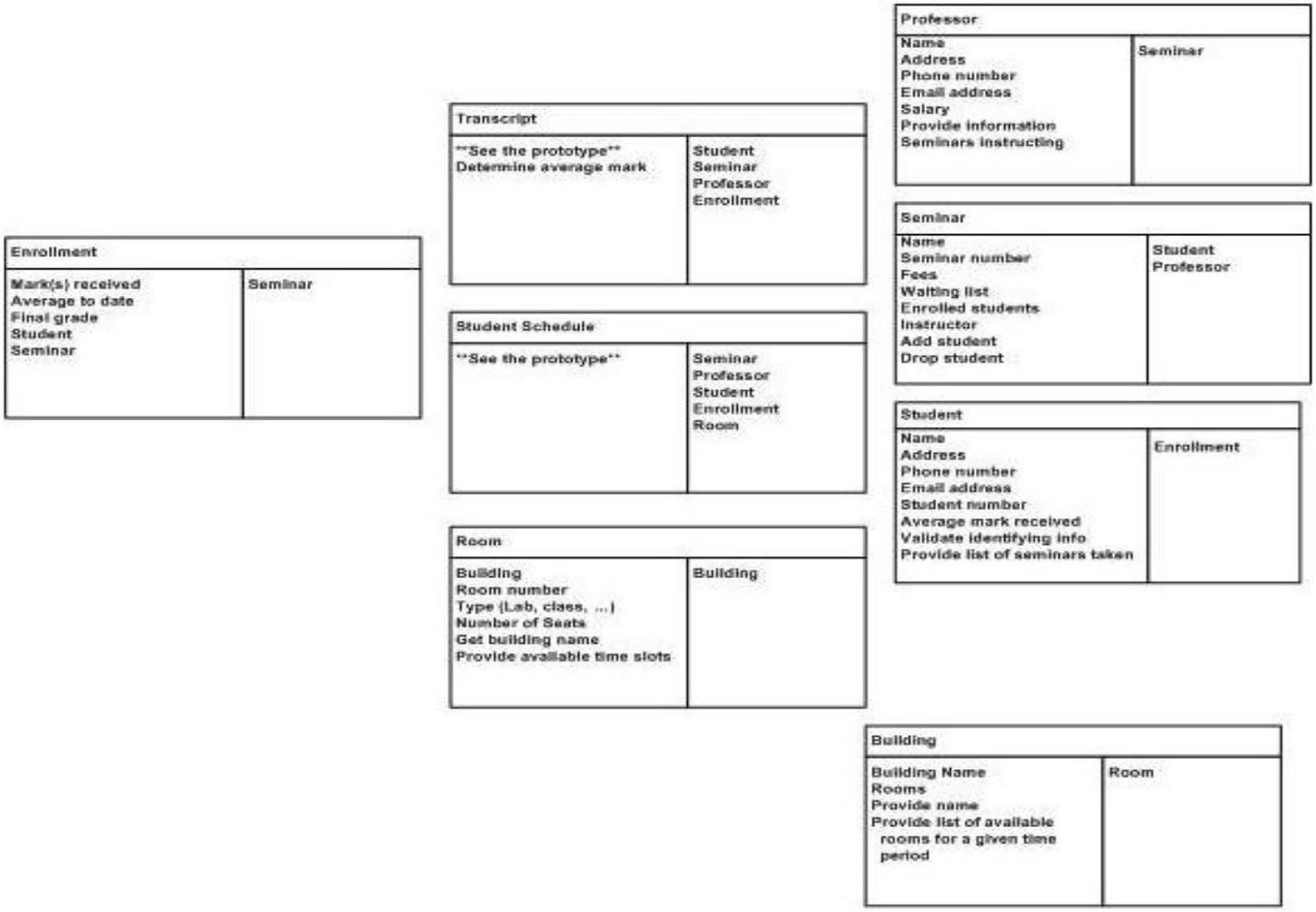

Order
Knows placement date
Knows delivery date
Knows total
Knows appl. cable taxes
Knows order number
Knows order items
Order Items

Student	
Student number Name Address Phone number Enroll in a seminar Drop a seminar Request transcripts	Seminar

Creation of CRC cards:

Iteratively perform the following:

- Find classes
- Find responsibilities
- Define collaborators
- Place the cards in proper order


Professor
Name
Address
Phone number
Email address
Salary
Provide information
Seminars instructing
Seminar
Transcript
See the prototype
Determine average mark
Student
Seminar
Professor
Enrollment
Seminar
Name
Seminar number
Fees
Waiting list
Enrolled students
Instructor
Add student
Drop student
Student
Professor
Enrollment
Mark(s) received
Average to date
Final grade
Student
Seminar
Seminar
Student Schedule
See the prototype
Seminar
Professor
Student
Enrollment
Room
Student
Name
Address
Phone number
Email address
Student number
Average mark received
Validate identifying info
Provide list of seminars taken
Enrollment
Room
Building
Room number
Type (Lab, class, ...)
Number of Seats
Get building name
Provide available time slots
Building
Building
Building Name
Rooms
Provide name
Provide list of available rooms for a given time period
Room

Advantages

- Simple: does not overwhelm the task of OO modeling.
- Emphasizes the division of responsibilities across objects
- Encourages an intuitive understanding of the model
- The size of the cards provides a guideline for class complexity
- The cards capture the essential aspects of the design: classes, responsibilities, and the patterns of interactions.
- Inexpensive, portable, flexible, and readily available

Disadvantages

- Problematic for large systems.
- There may be a large number of classes and maintaining a lot of CRC cards becomes difficult especially if there are many dependencies.

Resources

- http://www.agilemodeling.com/artifacts/crcModel.htm
- https://en.wikipedia.org/wiki/Class-responsibility-collaboration_card
- http://www.openloop.com/education/classes/umn_98/umn_cpp/crc.htm